The galaxy-bias parameter model of the lssty1xdesi forecast

A summary for collaborators — November 2026

1 Setting

The lssty1xdesi forecast combines LSST Y1 3x2pt with DESI spectroscopic clustering, measured as angular power spectra C_ℓ^{gg} in 76 thin redshift bins of equal comoving width, $\Delta d_{\rm com} = 62\,{\rm Mpc}$. The bins belong to six tracer groups, called chunks: BGS, LRG1–3 and ELG1–2. The shipped model carries one linear-bias amplitude per chunk, $b_1(z) = B_g/D(z)$, where b_1 is the Eulerian linear bias, B_g is the free amplitude of chunk g, and $D(z)$ is the linear growth factor normalized to $D(0) = 1$; seven Fingers-of-God velocity dispersions σ_v smear each bin's redshift distribution. This summary describes the planned replacement: two one-loop bias models built from the same bias operators, differing in what is free. **Model A applies at** $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$, the Fourier cut matched to the radial resolution of the thin bins. **Model B applies at** $k_{\rm max} \simeq 0.18\,h\,{\rm Mpc}^{-1}$, the shipped scale mask. Each model has a baseline and a pessimistic prior choice; Table 1 counts the parameters. The seven cosmological parameters and the seven σ_v are unchanged and not counted.

2 Linear bias: per bin, but not free per bin

The linear bias of thin bin i in chunk g is

$$b_{1,i} = \frac{B_g\,(1+\delta_i)}{D(z_i)}, \qquad \delta_i = s_g\,(z_i - \bar{z}_g) + \varepsilon_i, \tag{1}$$

where z_i is the bin redshift, $\bar{z}_g$ the mean redshift of the chunk's bins, δ_i the fractional deviation from the chunk trend B_g/D, s_g a tilt, and ε_i the residual around the tilted line. The amplitude B_g and the tilt s_g are free with flat priors, one each per chunk: 12 parameters. The residuals carry a correlated Gaussian prior within a chunk,

$$\langle \varepsilon_i\,\varepsilon_j \rangle = \sigma_g^2\,\exp\!\left(-\frac{|z_i - z_j|}{L}\right), \tag{2}$$

and zero across chunks; σ_g is the per-bin prior width and L is the correlation length in redshift. The baseline prior sets $\sigma_g = 3\%$ for BGS, sized to its flux-limit curvature, $\sigma_g = 1\%$ for the other chunks, and $L = 0.2$. The pessimistic prior sets $\sigma_g = 3\%$ for every chunk and $L = 0.05$: local selection features on the scale of the 4000 Å break crossing a filter edge, $\Delta z \approx 0.03$–0.05.

The tilt is free because even a conserved galaxy population departs from B_g/D by about 1% per $\Delta z = 0.05$ through the bias continuity equation [1], and a flux limit adds a larger smooth drift. The shape B_g/D is not passive evolution: it keeps the clustering amplitude $b_1(z)\,D(z)$ constant, while a conserved population obeys $b_1 - 1 \propto 1/D$.

3 Higher-order bias inherits the linear drift

The quadratic bias $b_{2,i}$ and the tidal bias $b_{s^2,i}$ of bin i follow the per-bin linear bias through calibrated relations, plus one free offset per chunk:

$$b_{2,i} = b_2^{\rm fit}(b_{1,i}) + \Delta_{2,g}, \qquad b_{s^2,i} = -\frac{4}{7}\,(b_{1,i} - 1) + \Delta_{s,g}, \tag{3}$$

with the separate-universe fit of Lazeyras et al. [2],

$$b_2^{\rm fit}(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3, \tag{4}$$

and the co-evolution relation for b_{s^2} [3], both in the CCL/FAST-PT normalization (the quadratic and tidal operators each carry a factor 1/2 in the galaxy overdensity). The trends take the full $b_{1,i}$ of Eq. (1), deviations included, so the higher-order sector drifts with the linear one. The third-order bias $b_{\rm 3nl}$ is fixed. The pessimistic variant adds a free tilt per chunk to each offset, $\Delta_g \to \Delta_g + t_g\,(z_i - \bar{z}_g)$, with t_g the offset tilt.

4 Counterterms and stochastic terms

The one-loop galaxy power spectrum in redshift space is, compactly,

$$P_{gg}(k,\mu) = P_{gg}^{\rm PT}(k,\mu) + \left(\alpha_0 + \alpha_2\,\mu^2\right) k^2\,P_{\rm lin}(k) + {\rm SN}_0 + {\rm SN}_2\,k^2\mu^2, \tag{5}$$

where k is the comoving wavenumber, μ the cosine between the wavevector and the line of sight, $P_{\rm lin}$ the linear matter power spectrum, α_0 and α_2 counterterm amplitudes, and $\rm SN_0$ and $\rm SN_2$ stochastic amplitudes. $P_{gg}^{\rm PT}$ is the one-loop spectrum of biased tracers: its tree level is the Kaiser spectrum $(b_1 + f\mu^2)^2 P_{\rm lin}$, with $f = d\ln D/d\ln a$ the linear growth rate, and its b_1^2 part carries the halofit nonlinear matter power. **Model A** keeps the real-space one-loop terms, linear Kaiser redshift-space distortions, the σ_v damping, one α_0 and one $\rm SN_0$ per chunk. **Model B** adds the radial counterterm $\alpha_2\,\mu^2k^2P_{\rm lin}$, the stochastic term $\rm SN_2\,k^2\mu^2$, and the one-loop redshift-space terms.

In the pessimistic cases α_0 and α_2 also receive a tilt per chunk, because the counterterm scale follows the host-halo population. $\rm SN_0$ and $\rm SN_2$ do not: the Poisson part $1/\bar{n}(z)$, with $\bar{n}$ the tracer number density, is known per bin and sits in the covariance, so the sampled amplitudes are deviations from Poisson. The priors are Gaussian, centered on zero. Model A: width 1.0 on $\Delta_{2,g}$, 0.5 on $\Delta_{s,g}$, $0.5/k^2_{\rm max} \simeq 89\,h^{-2}{\rm Mpc}^2$ on α_0 (a 50% correction to the linear spectrum at the cut), and twice the Poisson power on $\rm SN_0$. Model B: widths of order 2 and 1 on the offsets (the calibrations demoted to prior centers), the DESI DR1 width $12.5\,h^{-2}{\rm Mpc}^2$ on α_0 and α_2, twice the Poisson power on $\rm SN_0$, and the DESI DR1 width on $\rm SN_2$ [4].

5 The parameter budget

	A baseline	A pessimistic	B baseline	B pessimistic
scale cut $k_{\rm max}$ [h/Mpc]	0.075	0.075	≈0.18	≈0.18
linear bias, tilt per chunk (new)	6	6	6	6
linear bias, per-bin residuals held by the prior	64 (smooth)	64 (rough)	64 (smooth)	64 (rough)
b_2, b_{s^2} offsets	12	24 (+tilt)	12	24 (+tilt)
α_0	6	12 (+tilt)	6	12 (+tilt)
α_2	—	—	6	12 (+tilt)
$\rm SN_0$	6	6	6	6
$\rm SN_2$	—	—	6	6
new parameters	94	112	106	130
bias-sector parameters without a tight prior, total including the 6 existing amplitudes	36	54	48	72

Table 1: The bias-sector parameter budget of the four configurations. The 64 residuals are the 76 per-bin deviations minus the free amplitude and tilt directions of the six chunks; "smooth" and "rough" name the baseline and pessimistic kernels of Eq. (2). The last row counts the 6 existing amplitudes B_g, so it is not a subset of the "new" row.

For reference, the DESI DR1 full-shape analysis uses seven nuisance parameters per tracer chunk, 42 in total at $k_{\rm max} = 0.2\,h\,{\rm Mpc}^{-1}$ [4, 5], so model B baseline is that budget plus one b_1 tilt per chunk. Model B needs code before it can run: one-loop bias in the adjacent-bin non-Limber cross spectra, and one-loop redshift-space templates.

References

[1] J. N. Fry, "The Evolution of Bias", 1996, ApJ, 461, L65.

[2] T. Lazeyras, C. Wagner, T. Baldauf and F. Schmidt, *Precision measurement of the local bias of dark matter halos*, arXiv:1511.01096.

[3] S. Saito, T. Baldauf, Z. Vlah, U. Seljak, T. Okumura and P. McDonald, *Understanding higher-order nonlocal halo bias at large scales by combining the power spectrum with the bispectrum*, arXiv:1405.1447.

[4] DESI Collaboration, *DESI 2024 V: Full-shape galaxy clustering from galaxies and quasars*, arXiv:2411.12021.

[5] DESI Collaboration, *DESI 2024 VII: Cosmological constraints from the full-shape modeling of clustering measurements*, arXiv:2411.12022.